

Day 2 staged hints and answers — facilitator copy

Offer a first hint, then a second hint before showing the worked answer. Equivalent reasoning with correct conditions, units and amount basis is fine. The numbers are invented model outputs, not learner measurements or real Ni-Cu values. Keep this sheet separate from independent attempts.

D0 • Bridge

Hint 1: Day 1 had only A and two bulk phases. Name the new composition variable before naming an interface.

Hint 2: Ask what must be known at a boundary besides the surrounding bulk.

Answer: the new links need a binary bulk composition/model, then boundary site geometry, exchange or inventory rule, and candidate state energies. Only (a), the invented unary line crossing, was supported by Day 1. It neither supplies Ni-Cu parameters nor a boundary model.

D1 • Composition and balance

Hint 1: $n_B = z(n_A + n_B)$; the remaining amount is A.

Hint 2: Put $f_1 = 1 - f_2$ into $z = f_1 x_1 + f_2 x_2$.

Answer: (1) $n_B = 0.40$ mol and $n_A = 1.60$ mol. (2) $0.20 = (1 - f_2)0.10 + f_2(0.50)$ gives $f_2 = 0.25$ and $f_1 = 0.75$. B balance is $0.75(0.10) + 0.25(0.50) = 0.20$; A balance is $0.75(0.90) + 0.25(0.50) = 0.80$. These fractions sum to one. (3) $x = 0.20$ is B atom fraction **within ALPHA**. Area occupied by a phase requires geometry or an area measurement, and the phase amount fraction requires a phase-equilibrium/amount calculation. Neither is x .

D2 • Ideal bulk

Hint 1: Evaluate the reference as $-9000 + 12000x$ first.

Hint 2: The supplied $q(0.20)$ is dimensionless; multiply by RT with units J/mol.

Answer: (1) at $x = 0.20$, reference = -6600 J/mol atoms; the ideal mixing term is $8314.5q \approx -4160.596$ J/mol atoms, so $g_b \approx -10760.596$ J/mol atoms. The endmember weighted value omits the mixing contribution. (2) -10.50 kJ/mol is -10500 J/mol; it differs from -10502.902 J/mol by about 2.902 J/mol, consistent with rounding to 0.01 kJ/mol. The printed values do not show a distinct model.

D3 • Exchange

Hint 1: A fixed occupied site trades one A away for one B in.

Hint 2: Use $\ln(0.1/0.9) = -2.19722458$ and $\ln(0.2/0.8) = -1.38629436$.

Answer: (1) $-16144.844 - (-9876.020) = -6268.824$ J/mol atoms; $12000 + 8314.5(-2.19722458) \approx -6268.824$. Subtracting only μ_B would fail to account for the A atom replaced at a filled site. (2) at 0.20 , $\mu_B - \mu_A \approx +473.656$ J/mol atoms. A sign of an exchange slope is not an equality of two phase Gibbs energies or a two-phase global minimum. We allowed only ALPHA here.

D4 · Geometry and excess

Hint 1: Compute A separately in 8000 bulk and 200 boundary sites.

Hint 2: For atom/nm², use excess B count divided by **40** nm².

Answer: (1) bulk A = 8000(0.90) = 7200; boundary A = 200(0.85) = 170. With 800+30=830 B, A is 7370 and the total is 8200. (2) at $\theta = x_b$, reference and cell B counts agree and excess is zero. At $\theta = 0.08$, the boundary has 16 B, the bulk 800 B and the reference 820 B: excess = -4 B over 40 nm², or -0.10 atom/nm². There are still 816 B atoms in the cell; a negative **excess** means depletion relative to the chosen equal-site bulk reference.

Optional conversion: the worked 0.25 atom/nm² equals $0.25 \times 10^{18} / N_{Av} \approx 4.15135 \times 10^{-7}$ mol/m². The count/area route and $5(0.15 - 0.10)$ route agree. Using one boundary's area with both boundaries' count would double the reported value.

D5 · Open one-state selection

Hint 1: Multiply bulk odds 0.1/0.9 by 1.82459687.

Hint 2: If the resulting odds are r , solve $\theta = r/(1 + r)$.

Answer: (1) selected odds ≈ 0.20273299 , so $\theta_* \approx 0.16856026$. Since $\delta < 0$, B lowers boundary-site energy; the selected occupancy exceeds 0.10. At $\delta = 0$ the factor is one, $\theta_* = x_b = 0.10$, and this convention's excess vanishes. (2) θ is a **fraction of 200 sites**; expected boundary B count is about 33.7121 atoms, or $33.7121/N_{Av}$ mol, not 0.16856 mol. ϕ is a reservoir-subtracted excess grand potential per mole of boundary sites under a chosen reference, not absolute boundary free energy.

The excess is $5(0.16856026 - 0.10) = 0.3428013$ atom/nm². Converting with $10^{18}/N_{Av}$ gives 5.69235×10^{-7} mol/m². For both boundaries, excess B count ≈ 13.7121 over 40 nm².

D6 · Closed inventory

Hint 1: Total B stays 850. Boundary B is 200θ .

Hint 2: Divide the remaining B count by 8000 to obtain bulk x_b .

Answer: (1) at $\theta = 0.35$, boundary B=70, bulk B=780 and $x_b = 780/8000 = 0.0975$. Keeping 0.10 in bulk would give 800+70=870 B, contradicting the fixed 850. A remains 7350 because every B movement replaces an A movement in the opposite direction. (2) at the supplied minimum, boundary B ≈ 34.3215 and bulk B ≈ 815.6785 ; their sum is 850 within display rounding. State comparisons at different total B compare different systems, so an energy ordering for one closed inventory cannot be inferred from another. The closed θ_* need not equal the open 0.16856026 because the finite bulk fraction adjusts.

D7 · Candidate boundary states

Hint 1: Calculate $D = \text{II minus I}$ **within each table row**.

Hint 2: In row one, State II boundary B count is $200(0.2689826)$; subtract it from the shared 850.

Answer: (1) at $x_0 = 0.10$, $D = +22.21984$ J/mol all cell sites and I is lower. At $x_0 = 0.25$, $D = -5.47887$ J/mol all sites and II is lower. Different rows have different B totals (850 versus 2050) and are not a same-inventory comparison. The sign change brackets a crossing for the declared continuous uniform-state model; it does not by itself establish coexisting real boundary phases. (2) State II has boundary B ≈ 53.796516 , bulk B ≈ 796.203484 , and $x_{b,II} \approx 0.0995254$. State I has boundary B ≈ 34.3215 , bulk B ≈ 815.6785 , and

$x_{b,I} \approx 0.10195981$. Each sums to 850. Since the two selected occupancies differ, forcing a shared final x_b would change at least one state's total B. (3) omitting η_{II} defines a different model, so the printed energy would no longer follow from the stated function. The missing term is $200(2000)/8200 = 48.78049$ J/mol **all cell sites** at either row; the baseline is not 2000 J/mol all sites. It shifts State II's energy without changing its selected occupancy at fixed inventory.

D8 · What would make this a material case?

Hint 1: Ask whether a sentence describes invented arithmetic, a needed input, or an asserted observation/prediction about Ni-Cu.

Hint 2: Separate one-state segregation from a two-state transition.

Answer: (a) toy-model result; (b) missing evidence, bulk model; (c) unsupported real claim; (d) missing evidence, boundary model, including compatible states under one reference. If a real single state were validated along with the exact case and bulk, a **conditional segregation** question could then be calculated, with its own checks. A structural transition additionally needs at least two matched real state functions, physical interface coexistence conditions and independent evidence. A toy energy crossing or solver return is not such evidence. A crossing compares only *uniform* states: at fixed inventory, a boundary partly in I and partly in II can have lower total energy than either uniform state near the crossing, so the change may happen over a range of compositions rather than as a sharp switch. Only a coexistence analysis decides this.

D9 · Fresh integrated card

Hint 1: Open odds start at $0.20/0.80 = 0.25$; closed B starts from $8000x_0 + 200\theta_0$.

Hint 2: For excess use $5(\theta_* - x_b)$ atom/nm². For the closed trial, subtract boundary B from fixed total B.

Answer: (1) open odds = $0.25(1.82459687) = 0.45614922$ and $\theta_* \approx 0.31325719$. Excess $\$ = 5(0.31325719 - 0.20) \approx 0.566286$ atom/nm². Across 40 nm², excess B count is $0.566286(40) \approx 22.6514$ atoms, also $200(0.31325719 - 0.20) \approx 22.6514$. (2) closed total B = $8000(0.20) + 200(0.20) = 1640$. At $\theta = 0.40$, boundary B=80, bulk B=1560 and $x_b = 0.195$. Keeping 0.20 would give 1680 B, violating the 1640-B inventory. (3) Example: "The invented uniform-state model orders I below II at 850 B and II below I at 2050 B." "It supplies no validated Ni-Cu bulk and matched boundary-state energies or physical interface coexistence analysis, so neither a measured transition nor a strengthening effect follows." Other precise supported/gap pairs are acceptable.

D10 · Supported exit

Hint 1: Day 1 gives the potential choice; D5 and D6 give different matter-exchange boundaries.

Hint 2: Reuse D5's zero-preference limit and D7's first row.

Answer: (1) minimize total G for the closed cell at fixed T, p and fixed A/B counts. In the open boundary experiment the combined cell+reservoir conserves atoms while the cell exchanges B for A at fixed occupied boundary sites and fixed reservoir μ_A, μ_B . In the finite closed cell, both cell A and B totals are fixed and bulk x_b changes. (2) $\theta_* = 0.10$ and $\Gamma_B = 0$ by the chosen equal-site convention. The zero follows from an invented zero-preference model, not observed segregation data. (3) $D = +22.21984$ J/mol all sites; I is lower among the two uniform candidates. Examples of missing conditions: compatible real state free energies and geometry, physical interface coexistence, uncertainty/held-out

evidence, or demonstrated stability against other candidates. (4) The named next task or lesson should match the learner's question (for example D1-D3 → Task 01, D4-D6 → Task 04, D7 → Task 03 or 04); neither primer validates a Ni-Cu prediction.

Facilitator interpretation

Do not turn this answer sheet into a pass/fail threshold. Record whether the learner needed the printed factor, partner help or supplied state energies, and which *concept* needs more practice: binary balance, energy basis, exchange, area/excess, ensemble or real-claim evidence. Point that need to the matching worked example (Tasks 00-05) or optional detailed lesson, not automatically to the entire lesson library.